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# The CoRA-AI Assistant: A Multi-Agent Architecture for Analysing Conflict and Crisis Data

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## Abstract

Policy analysts working on conflict early warning, prevention, and crisis response increasingly rely on heterogeneous data sources. The European Commission's Joint Research Centre (JRC) already produces conflict forecasting models, trend analyses and data briefs that support early warning, but building on these outputs in applied work remains challenging. Analysts work under time constraints and often lack the technical expertise required to analyse structured and unstructured data and draw insights from them. This report presents the Conflict Risk Analysis (CoRA) AI Assistant prototype, a multi-agent system based on large language models (LLMs) that complements the JRC's existing conflict early warning tools. By enabling natural-language interaction with integrated datasets offering global coverage and daily to monthly updates, the CoRA-AI Assistant aims to lower the technical barrier to working with conflict data. The system orchestrates specialised components for query classification, code generation, document retrieval, and synthesis, allowing users to produce outputs that combine quantitative results with qualitative context. The project team evaluated the system through user testing, interviews, surveys, and written feedback. Overall, users perceived the CoRA-AI Assistant as a useful first-pass analytical and briefing tool, while also identifying limitations in handling complex queries, time-related queries, and multi-step interactions. Some of these limitations may be related to the underlying LLM architecture and system configuration used during testing, although the evaluation does not establish a direct causal relationship. The report outlines priorities for further development, including usability, analytical robustness, and data coverage.

**Keywords:** large language models, multi-agent systems, conflict analysis, climate–conflict nexus, retrieval-augmented generation, natural-language data querying, computational social science

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# 1 Introduction

Policy analysts working on conflict prevention and crisis response increasingly rely on heterogeneous data sources, including conflict event datasets, climate indicators, humanitarian risk indices, model outputs and unstructured reports. At the European Commission's Joint Research Centre (JRC), the Conflict Risk Analytics Team in the Defence and External Security Unit (JRC.E7) develops and maintains conflict monitoring and forecasting tools that draw on such sources. Its outputs, including conflict forecasting models, trend analyses and data briefs, support the European Union's conflict early warning system and broader policy processes related to conflict prevention and crisis management. These activities are undertaken in collaboration with the European External Action Service (EEAS), with the support of the Service for Foreign Policy Instruments (FPI), and are embedded in the European Union's updated framework for Early Warning, Conflict Analysis and Conflict Sensitivity. [1]

In this context, the team was tasked with developing an AI Assistant that allows policy analysts to rapidly synthesise information from multiple sources. These currently include event-level datasets such as the Armed Conflict Location and Event Data Project (ACLED) (Raleigh et al., 2010) and the Uppsala Conflict Data Program Georeferenced Event Dataset (UCDP GED) (Sundberg and Melander, 2013), risk indices such as the INFORM Risk Index (INFORM) (INFORM, 2024), climate data from the Climate Change Vulnerability Index (CCVI) (Mittermaier et al., 2026), and unstructured sources including humanitarian reports and field-based analyses. Together, these sources provide insight into patterns of violence, vulnerability, and environmental stress.

Despite the growing availability of data and analytical tools, translating these inputs into timely and actionable insights remains difficult. Policy makers and analysts work under time constraints and often lack the technical expertise required to combine structured datasets, geospatial information, and unstructured evidence in their daily workflows. Doing so typically requires programming and data analysis skills, creating a bottleneck that limits the use of available data in policy-relevant contexts.

Recent advances in artificial intelligence (AI), particularly large language models (LLMs), offer new ways to address this gap. LLMs can translate natural language into executable code (Chen et al., 2021; Li et al., 2024) and retrieve relevant information from large document collections (Lewis et al., 2020). This has led to the development of "data assistants" that let users interact with complex data systems through conversational interfaces (Rajkumar et al., 2022; Liu et al., 2023). In crisis analysis, such systems could widen access to conflict data, allowing policy users to formulate analytical queries in natural language without relying on specialised know-how.

Existing applications in social science, early warning and crisis management have focused primarily on qualitative tasks, such as document classification, sentiment analysis, and retrieval-augmented generation (RAG) over policy or news corpora (Ornstein et al., 2024; Bail, 2024). These systems function as advanced search and summarisation tools but provide limited support for structured, quantitative analysis. Jiménez et al. (2025), for example, propose a GraphRAG-based system that combines conflict data and humanitarian reporting within a knowledge graph framework, improving the robustness of document-based approaches. However, existing approaches remain largely document-centred, with limited integration of quantitative data.

This matters for policy applications, where quantitative analysis plays a key role in understanding conflict dynamics and supporting evidence-based decision making. Working with structured datasets, such as conflict events or spatial indicators, requires programming expertise that many users do not have. Frameworks such as PRIO-GRID (Peace Research Institute Oslo; Tollefsen et al., 2012) and

[1] https://www.eeas.europa.eu/eeas/conflict-prevention-peace-building-and-mediation_en

aggregated indicators such as CCVI support data harmonisation, but their use in integrated analysis still remains complex.

Multi-agent systems offer one way to address these challenges. By splitting complex tasks into components that are handled by cooperating agents, such systems can support a wider range of analyses and adapt more flexibly to user needs (Wu et al., 2023; Hong et al., 2023; Qian et al., 2023). In data analysis, separate agents can handle code generation, document retrieval, and result validation, capabilities that are difficult to achieve in single-agent AI Assistant systems.

This report introduces the Conflict Risk Analysis (CoRA) AI Assistant prototype, a multi-agent system built on this principle and designed to complement the JRC's existing conflict early warning tools. The CoRA-AI Assistant provides a natural-language interface to integrated datasets that offer global coverage and daily to monthly updates, alongside a corpus of humanitarian reports and field-based analyses. Behind this interface, the system orchestrates specialised components for query classification, code generation, document retrieval, and synthesis. A user can therefore ask for conflict event counts, spatial patterns, or contextual background without writing code, and receive outputs that pair quantitative results with qualitative context.

The project team evaluated the prototype through user testing, interviews, and surveys, and written feedback with JRC staff. Users describe the CoRA-AI Assistant as a valuable first-pass analytical and briefing tool, but they also report limitations in handling complex queries, time-related queries, and multi-step interactions. Some of these limitations are linked to the underlying LLM architecture used during testing, and the report sets out priorities for addressing them.

The remainder of this report is organised as follows. Section 2 describes the integrated data sources and presents the system architecture. Section 3 reports testing results and user feedback. Section 4 discusses the AI Assistant's strengths, limitations and next steps, and Section 5 concludes the report.

## 2 Data and System Architecture

This section describes how the CoRA-AI Assistant integrates data sources and processing components to support conflict analysis through a single interface. The design offers a user-friendly way to query and combine large and heterogeneous datasets.

### 2.1 Integrated data sources

The CoRA-AI Assistant currently brings together 7 datasets within a central PostgreSQL database, using the `pgvector` extension to support semantic search over unstructured text documents. Table 1 summarises the main data sources integrated into the system.

**Table 1.** Integrated data sources in the CoRA-AI Assistant prototype.

| Dataset | Rows | Coverage | Description |
|---|---|---|---|
| ACLED Events | ~3 000 000 | 1997–Present | Armed conflict events in 238 countries and territories: battles, violence against civilians, protests, riots, explosions, strategic developments |
| UCDP GED | ~420 000 | 1989–Present | Georeferenced armed-conflict events from 132 countries with fatality estimates |
| PRIO-GRID Climate | ~58 000 000 | 1951–Present | Monthly gridded climate indicators: drought index (SPEI-12), heatwaves, population density |
| INFORM Severity | 5765 | Present | Crisis severity indicators covering humanitarian conditions and trends |
| GADM Boundaries | 3919 | Present | Administrative boundaries for spatial analysis and map generation |
| PRIO Base Grid | 63 606 | Static | Reference grid for aligning climate and conflict data |
| RAG Documents | ~315 000 | 1996–Present | Embedded document corpus: ReliefWeb reports and UCDP encyclopedia entries |

*Note:* Row counts are approximate and reflect the state of the data at the time of writing. ACLED is updated weekly, UCDP GED monthly, and the document corpus continuously, so these figures grow over time. Counts for GADM and PRIO Base Grid are fixed.

### 2.2 User interaction

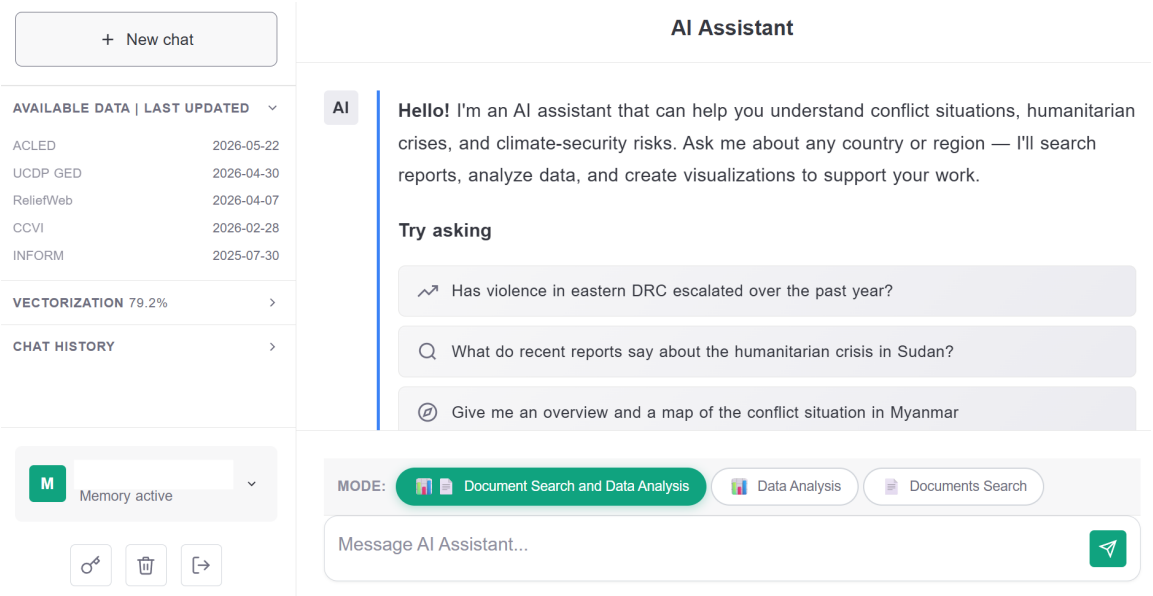
Users interact with the system through an AI Assistant interface that allows them to query, analyse and visualise data in natural language. Figure 1 shows the main interface, and Figures 2 and 3 give examples of the system's output in response to two user queries.

The system is designed to support analytical tasks policy users frequently carry out. Users can ask questions such as:

- *Has violence in a given country or region escalated over the past year?*
- *What do recent reports say about the humanitarian or conflict situation in a specific context?*
- *Provide an overview and a map of the conflict or environmental situation in a given country or region.*

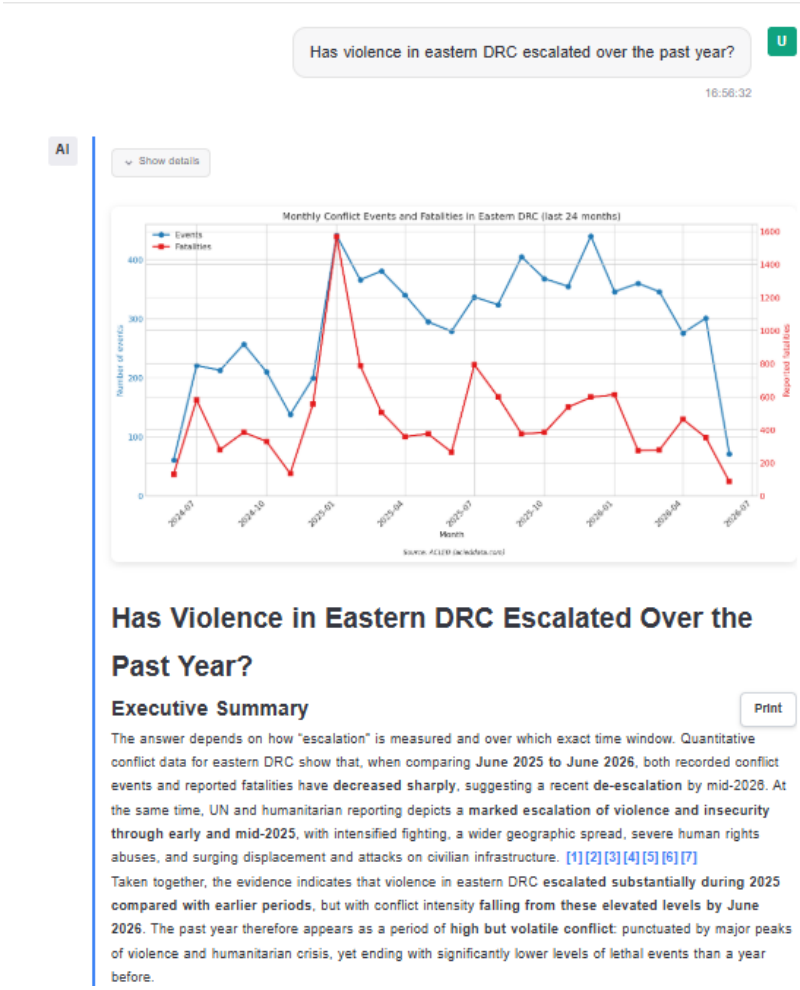
Depending on the query, responses can include text summaries grounded in sources with citations, charts, and maps, along with an interpretation of the underlying data and figures. Users can also export the underlying results, so that tables and figures can be reused outside the system.

**Figure 1.** User interface of the CoRA-AI Assistant prototype.



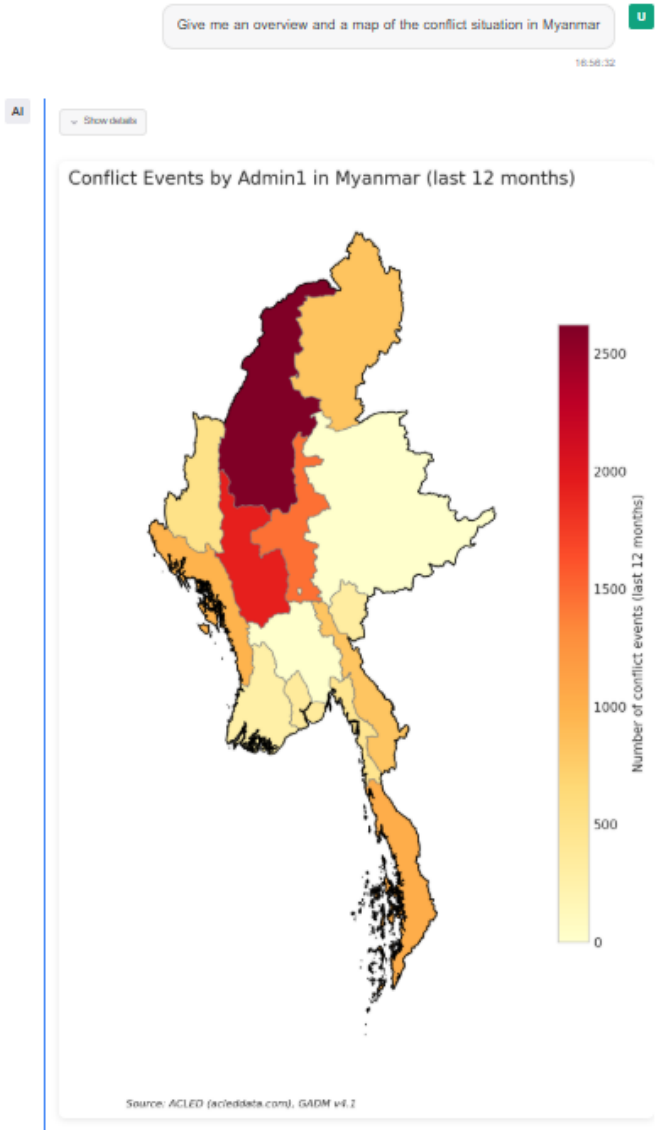

*Source:* CoRA-AI Assistant prototype, JRC analysis.

**Figure 2.** Example query on violence in the Democratic Republic of the Congo (DRC) and the corresponding output, including a graph and a summary.




*Source:* CoRA-AI Assistant prototype, JRC analysis.

**Figure 3.** Example query on the conflict situation in Myanmar and the corresponding output, including a choropleth map.

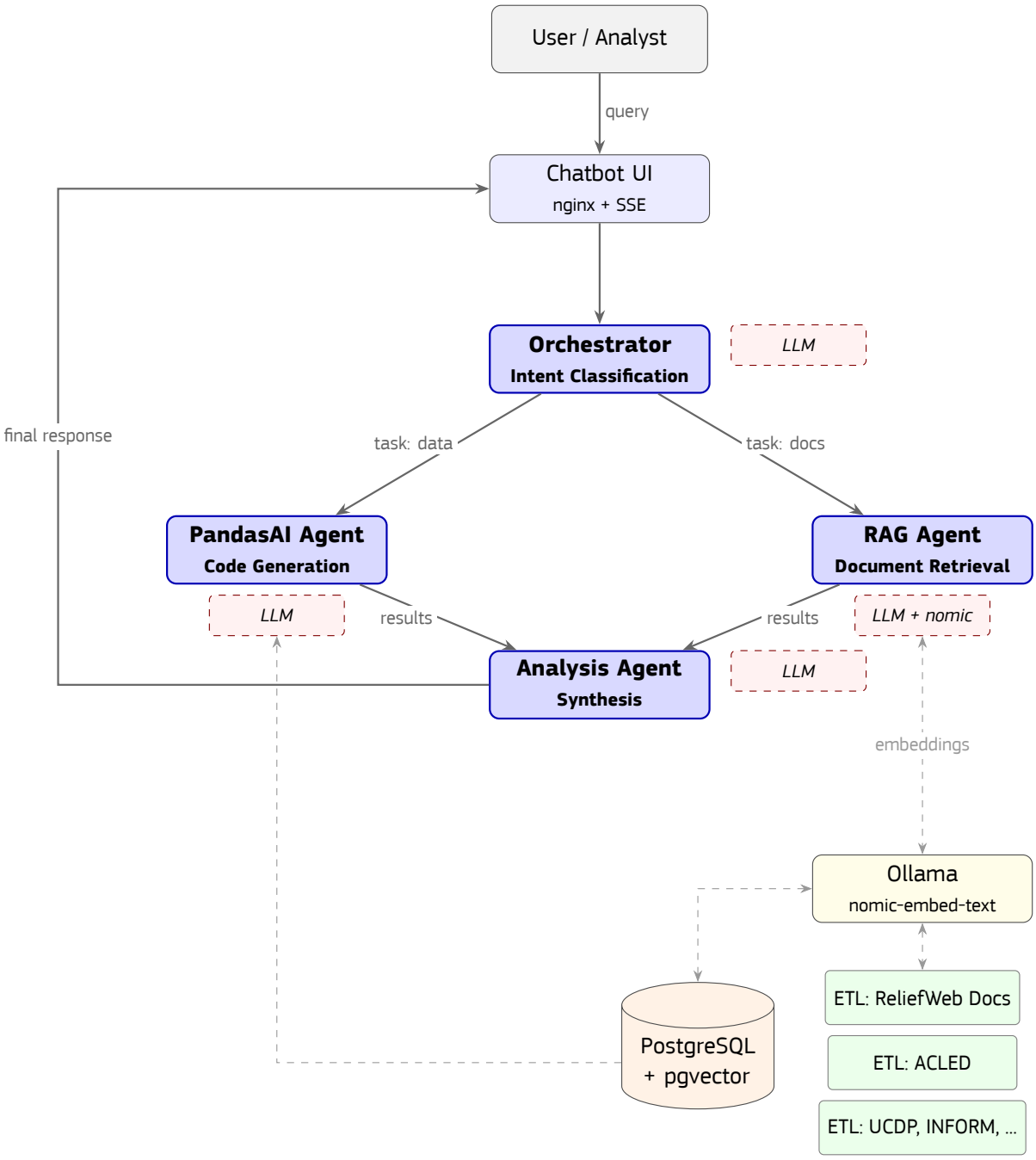



*Source:* CoRA-AI Assistant prototype, JRC analysis.

## 2.3 System architecture

At a high level, the CoRA-AI Assistant functions as a modular system made up of specialised agents. Users submit a query through the interface and can specify whether they need a data analysis, document search, or both. The system interprets the request, routes it to the relevant agents, and combines their outputs into a single response. Figure 4 provides an overview of this process.

**Figure 4.** System architecture of the CoRA-AI Assistant prototype.




*Note:* Solid arrows indicate processing flows; dashed arrows indicate embedding operations. *Source:* JRC analysis.

### 2.3.1 Multi-agent design

The AI Assistant functions as a multi-agent system in which each agent is responsible for a specific task, such as information retrieval, data analysis, or response planning and orchestration. Each agent consists of an LLM, a system prompt that provides a stable set of background instructions, and access to the tools it needs, such as a database connection or a code execution environment. The system uses an orchestrator agent to define the workflow and decision-making process, after which other agents generate specific outputs that are then combined into the final response.

The system is designed to work with any LLM accessible through an application programming interface (API), including in-house models such as GENESIS or EC-GPT (the default), commercial models such as those from OpenAI and Anthropic, and open-source or self-hosted models. In principle, users with access to different providers can therefore choose the model that best matches their requirements in terms of performance, cost, speed, data governance, and security policy.[1] However, the performance of the system as a whole, including response quality, reasoning, latency, and cost, depends strongly on the model selected.

### 2.3.2 Query interpretation and routing

The first step in the process is interpreting the user query. Users can select one of three analysis modes through the AI Assistant interface: (i) document search and data analysis (the default), (ii) data analysis only, and (iii) document search only. This selection limits which agents may be used.

Within these constraints, the agents decide what output they can produce. If a query calls for data analysis but no relevant data exist, for instance, the data analysis agent can decline to carry out the request rather than return an unsupported result.

Earlier versions of the system let the orchestrator classify queries without specific user input. This proved unreliable, a queries were often misrouted, and the system prompts needed to correct errors were long, with detailed examples of specific query types and how to respond. Letting users select the analysis mode narrowed the orchestrator's task and made routing more reliable. As with prompt design more generally, we expect that more capable models will reduce the need for both detailed instructions and explicit user input.

### 2.3.3 Data analysis

When a query requires data analysis, the CoRA-AI Assistant uses a code-generation agent to translate the user's request into executable queries over the underlying datasets. This is done within a Python environment combined with structured query language (SQL) queries. The system relies on semantically annotated datasets, which include descriptions of tables, variables, and valid values. This allows the model to interpret user requests such as "violent events" or "civilian targeting" and map them to the relevant data fields. The outputs of this agent include tables, statistical summaries, charts, and maps, depending on the query.

Because responses are grounded in data and generated from query results rather than from the model's parametric knowledge, this approach reduces the risk of hallucination. The interface further supports this by allowing users to inspect the reasoning steps and the code the agent generated, so that a reported figure can be traced back to the query that produced it.

Depending on deployment requirements, the system can also be configured to transmit only dataset schemas and metadata to the LLM, while running all data retrieval and analysis code locally. In this configuration the data itself never leaves the system environment. This is currently implemented through a "secure mode" option in the user interface.

[1] System prompts are not fully portable across models. More capable models tend to perform better with fewer instructions, while less capable ones require more explicit rules. Switching models therefore typically requires some prompt adjustment.

### 2.3.4 Document search

For queries that require contextual information, the system uses a retrieval-augmented generation (RAG) approach to search a corpus of documents, including humanitarian reports and encyclopedic sources. Rather than matching keywords, the system compares numerical representations of language (text embeddings), generated by a locally hosted embedding service (Ollama, `nomic-embed-text`). This allows it to retrieve relevant passages even when the query and the document use different vocabulary.

We designed the RAG system to address two challenges. First, short user queries do not always align well with the long, detailed documents that might answer them. Second, semantic similarity also does not guarantee that a passage is useful: a report on a different country, region, or period may match closely on language while being wrong on context. The retrieval process therefore proceeds in four steps:

**Query expansion**

The system first uses an LLM to generate a hypothetical document relevant to the user's query, and embeds that document rather than the query itself (Gao et al., 2023). Because the generated text resembles the documents being searched in language and structure, its embedding sits closer to genuinely relevant passages than the embedding of the original query would. The hypothetical answer does not need to be factually correct, as it is never shown to the user and serves only to locate relevant passages. In testing, we found that this step improved overall retrieval quality.

**Semantic search**

The embedded hypothetical document is compared against pre-processed document segments in the database, returning an initial set of candidate passages ranked by similarity.

**Recency weighting**

Candidates are re-weighted by publication date, so that recent reporting is favoured over older material covering the same topic. This reflects the requirements of early warning work, where current information usually matters more than historical background.

**Relevance filtering**

A second LLM assesses each remaining candidate against the original user query and discards those that are not relevant. This step catches the nuances that similarity scores miss, in particular passages that match on subject matter but refer to the wrong country, region, or time period. Only the passages that survive filtering are passed to the synthesis agent.

### 2.3.5 Combined analysis and synthesis

The CoRA-AI Assistant can combine structured data analysis with document retrieval within a single query. When both types of information are required, the system processes them in parallel and merges the results into a single output.

The synthesis agent is designed to interpret quantitative results, draw links to supporting evidence from the retrieved documents, and produce a structured response with references. Where the data analysis agent has produced a chart or a map, the synthesis agent inspects the figure and describes what it shows, so that analysts receive an interpretation of the visual output alongside the figure itself.

## 2.4 Data pipelines

The datasets described in Section 2.1 are kept up-to-date by automated extract, transform, and load (ETL) pipelines. These retrieve new records from ACLED, UCDP, INFORM, and ReliefWeb, harmonise them against the schemas held in the database, and write them to PostgreSQL. Documents retrieved from ReliefWeb are additionally embedded and indexed for semantic search. Pipelines run on a schedule that matches the update frequency of each source, so that the system reflects the most recent data available without manual intervention.

# 3 Testing and User Feedback

## 3.1 Test setup

The testing phase aimed to assess the usability, analytical performance, and perceived value of the CoRA-AI Assistant prototype, and to identify its main limitations and priorities for further development. The evaluation focused on perceived usability, accuracy and reliability, data and visualisation performance, workflow integration, and barriers to adoption.

Testing was conducted over one week in April 2026 with ten staff members of the European Commission's JRC. Access was limited to JRC staff for security reasons, and the evaluation ran through three complementary channels:

— five semi-structured interviews (30-minute sessions);

— three responses to an internal survey;

— two users providing written feedback after hands-on testing.

Participants included policy analysts, conflict and migration specialists, and technical users with coding and data analysis experience, as well as less frequent users of AI tools. Most participants were regular users of general-purpose LLMs such as ChatGPT, Claude, and Gemini, so they often compared the system to those tools, for example on conversational fluency, handling of follow-up queries, output formatting, and interface design.

The system was run with GPT-5.4-mini during testing, the most suitable option within existing constraints. Practical constraints during the testing period ruled out the use of more capable models, which are likely to perform better.

## 3.2 User feedback

### 3.2.1 Overall assessment

Across all feedback channels, the CoRA-AI Assistant prototype was consistently seen as a promising tool, particularly for conflict analysis and briefing preparation. Users identified its main strength as a first-pass analytical assistant, able to synthesise information across multiple data sources quickly.

The system was seen as less effective in more complex interactions. Performance was weaker when queries required sustained conversational context, iterative refinement, or precise interpretation of user intent. Users nevertheless valued the system's transparency, particularly its explicit acknowledgement of its own analytical limitations.

The system performs best when queries are clearly defined, focused on specific datasets, and aimed at descriptive summaries or structured outputs. Performance declines when queries depend on incomplete or missing data, require high geographic precision, involve several follow-up interactions, or depend on strict temporal constraints. Table 2 gives examples of tested queries and how the AI Assistant performed on each.

**Table 2.** Tested queries and observed performance.

| Source | Query | Mode | Performance | Strengths | Limitations |
|---|---|---|---|---|---|
| Interview 1 | Default and exploratory queries | Data + Documents | Good | Accurate responses; useful sources; time-saving | Limited follow-up handling; no export functionality |
| Interview 2 | Greenland tensions | Data + Documents | Poor | – | Data unavailable; lack of explicit explanation |
| Interview 2 | Conflict time series (Lebanon, Mexico) | Data analysis | Mixed | Correct visualisation; follow-up handled | Output overly verbose |
| Interview 2 | Drought map (Europe, NUTS2) | Data analysis | Poor | Attempted mapping | Inconsistent administrative units |
| Interview 3 | Iran–Lebanon conflict impacts | Data + Documents | Mixed | Coherent narrative; relevant sources | Limited map clarity; dataset dominance |
| Interview 3 | MENA outlook (6 months) | Data + Documents | Poor | Structured output | Time frame not respected; outdated sources |
| Interview 3 | Humanitarian impacts (Middle East) | Data + Documents | Mixed | Clear structure | Weak temporal handling; unclear charts |
| Interview 3 | Drought-conflict hotspots | Data + Documents | Mixed | Partial answer | Missing datasets |
| Interview 4 | North Korea escalation | Data + Documents | Mixed | Useful framing; charts | Key aspects omitted; default time window unclear |
| Interview 4 | Nuclear arsenal follow-up | Documents | Good | Transparent limitations | Limited source coverage |
| Interview 4 | Iran protests (2026) | Data + Documents | Good | Useful visualisation | Limited analytical depth |
| Interview 4 | Localisation follow-up | Follow-up | Poor | – | Insufficient spatial precision |
| Interview 5 | Serbia protests | Data + Documents | Good | Accurate; briefing-ready output | – |
| Interview 5 | Summary follow-up | Follow-up | Poor | – | Loss of conversational context |
| Interview 5 | Deadliest events graph | Data + Documents | Good | Successful visualisation | – |
| Interview 5 | Graph-to-map conversion | Follow-up | Poor | – | Context loss |
| Survey | Iran–Lebanon impacts | Not specified | Mixed | Accurate text analysis | Visual inconsistency |
| Survey | PNG conflict analysis | Documents | Good | Structured output | Missing expected sources |
| Survey | Azerbaijan outlook | Not specified | Poor | – | System error |
| Survey | Guatemala fatalities | Data + Documents | Good | Accurate results | Minor clarity issues |
| Email | China–Japan dispute | Not specified | Poor | – | Incorrect interpretation |
| Email | Croatia–Serbia dispute | Follow-up | Poor | – | Not covered by datasets |
| Email | Reformatting request | Reformatting | Poor | – | Re-analysis instead of formatting |
| Email | General query | Not specified | Mixed | Relevant output | Visualisation failure |

### 3.2.2 Interface usability

Users reported a positive first experience with the AI Assistant UI, which was generally seen as intuitive and visually clear. Most were able to interact without difficulty, and the interface's example queries helped guide initial exploration.

Usability problems did arise during longer sessions. The interface was often seen as too dense, with too many elements visible at once, including background processing information and multiple panels. Users consistently preferred a simpler default layout, with further detail available on demand.

Several modes and labels were also unclear. The differences between analysis modes (data analysis, document search) were not always obvious, and some options were not well understood. Technical terms and system indicators were frequently misread, pointing to a need for clearer labelling and guidance.

Most of this feedback has already been addressed, with a simplified layout, background information hidden by default, and clearer labels for the different configurations. Additional improvements, for example around terminology and background information, will be informed by further rounds of user feedback.

### 3.2.3 Accuracy and data limitations

The main weaknesses identified concern the accuracy of outputs and the handling of data limitations. Users frequently encountered outputs based on outdated or incomplete information, particularly when queries required recent or forward-looking analysis, and temporal queries were not always carried out correctly.

Where data coverage was incomplete, the system did not always signal this. Outputs that appeared complete despite partial or missing data reduced user trust. Feedback suggests that users tolerate data gaps well when these are stated explicitly.

Maintaining user intent across multi-step interactions was a further problem. The system sometimes failed to incorporate clarifications or refinements, producing responses that were disconnected from earlier context. Users found iterative interactions weaker than in general-purpose LLMs.

Some of these issues can be addressed directly: the code generation agent's instructions will be revised to give it better knowledge of the available datasets, handle temporal queries more reliably, and flag data limitations more clearly. Multi-step interactions are harder. We plan to limit the number of follow-up interactions, both to keep queries well-defined and to control token costs; where they do occur, a more advanced reasoning model for the orchestrator would likely help.

### 3.2.4 Visual outputs

Maps and charts were identified as a key strength, with clear potential for briefing and reporting workflows. They were also a frequent source of frustration when not implemented well.

Common issues included insufficient zoom levels, missing legends, unclear variable definitions, and inconsistencies between graphical outputs and the accompanying text. Comparative visualisations were particularly sensitive to inconsistencies in geographic units.

When visualisations could not be generated, users expected a clear explanation. Without one, it was often unclear whether the cause was a data limitation, rate limiting, or other system behaviour.

While developing the prototype, we noticed that the quality of visual outputs depends strongly on the model used for the code generation agent: more advanced LLMs produce more polished figures and make fewer coding errors. The agent's system prompts also affect output quality. We will explore both factors in the coming months and expect outputs to improve as more capable models become available.

### 3.2.5 Workflow integration and output reuse

User behaviour during testing suggests the AI Assistant is likely to be used not only for analysis but also as a drafting and briefing tool. This revealed limitations in workflow integration: requests to reformat existing outputs were often interpreted as requests for new analysis.

Users wanted clearer separation between analytical queries and formatting or summarisation requests. There was also demand for better output reuse, including copy functionality, export options, and structured referencing.

Users further emphasised the importance of broader and more transparent source coverage. Beyond the core datasets, they expected the integration of additional institutional sources and clearer communication of what the data does and does not cover.

We will address these needs in several ways. The orchestrator will be updated so that requests to reformat or reuse existing outputs are handled separately from new analytical queries, without rerunning the analysis. Output reuse, such as copying text or exporting figures and references, will be added as frontend features rather than new agents. Broader source coverage will be addressed by adding retrieval sources, with the synthesis step stating clearly what each response covers.

### 3.2.6 Priorities from testing

Testing pointed to a consistent set of priorities for the next phase: separating reformatting from analytical queries, improving the system's awareness of its own data boundaries, adding output reuse features, and broadening source coverage. The high level of engagement, including voluntary follow-up feedback, suggests the tool's potential.

# 4 Discussion and Next Steps

This chapter reflects on key lessons learned from the AI Assistant prototype, drawing on the evaluation in Chapter 3. It first sets out the system's main strengths, then the limitations identified during testing and the development priorities that follow from them.

## 4.1 Strengths and contributions

The CoRA-AI Assistant demonstrates how LLM-based multi-agent systems can lower the technical barrier to data-driven conflict analysis. It enables users without programming expertise to run cross-dataset queries that would otherwise require advanced programming and data analysis skills. By combining schema-driven code generation, retrieval-augmented document search, and multi-agent orchestration, it produces both quantitative analyses and contextual insights through natural-language interaction.

A central strength is the integration of multiple data types and sources. Combining structured datasets with document corpora such as humanitarian reports lets the CoRA-AI Assistant produce more grounded and context-relevant outputs than approaches relying on a single data type.

The system also supports secure deployment configurations, in which only database schemas are sent to the LLM while computations run locally. This allows LLM-based analytical tools to be used where data sensitivity, governance, and compliance matter, while being adaptable to different use cases.

## 4.2 Limitations and next steps

The evaluation highlighted several areas for further development.

First, analytical performance depends on the underlying LLM, particularly for complex queries requiring multi-step reasoning, context retention, and code generation. Although the architecture is model-agnostic, testing relied on GPT-5.4-mini; more capable models are expected to improve analytical robustness and interaction quality as they become more accessible.

Second, user interaction and workflow integration need strengthening. Future iterations should simplify the interface, clarify the distinction between analytical and formatting requests, and support output reuse through export and reference handling.

Third, transparency remains a priority. The system should more consistently flag cases where data are incomplete, outdated, or outside its coverage, and communicate the temporal assumptions applied during analysis. Better context retention across multi-step interactions will also support more complex workflows.

Finally, visual outputs and data coverage require refinement. Maps and charts need clearer contextual information, including legends, consistent geographic units, and variable definitions. Expanding the range of validated data sources, in collaboration with domain experts, will further strengthen the tool's reliability and relevance.

# 5 Conclusion

This technical report presented the CoRA-AI Assistant prototype, a multi-agent system designed to support conflict analysis by integrating 7 heterogeneous datasets, including conflict event data, climate indicators, and humanitarian reports, into a single natural-language interface. The system combines schema-driven code generation, retrieval-augmented document search, and multi-agent orchestration to let non-technical users perform cross-domain analyses that would otherwise require advanced programming skills. The resulting software is operationally mature and ready for deployment, providing a functional basis for integration into an analyst-facing environment.

User feedback shows that such systems can lower the technical barrier to data-driven analysis, narrowing the gap between policy analysts and complex data infrastructures. Users can formulate queries in natural language and obtain both quantitative results and qualitative context within a single interaction, which is particularly relevant where analysts must rapidly synthesise multiple sources into timely, evidence-based briefings. Testing showed the prototype is most valuable for well-defined analytical tasks and briefing preparation. It also showed that accuracy and user trust depend strongly on how clearly the system communicates its data limitations and behaviour.

These limitations are partly linked to the model used during testing, GPT-5.4-mini, which was chosen as a cost-efficient and operationally feasible option. More advanced models could improve analytical quality and interaction robustness, subject to cost, access, and deployment considerations. Importantly, these considerations do not prevent deployment of the current system, but rather define opportunities for further optimisation and enhancement.

Future development will address the main limitations identified during testing: accuracy of responses, interaction capabilities, visual outputs, ease of use, and the coverage of integrated data sources. Progress will also depend on continued improvements in the availability and affordability of capable LLMs.

More broadly, this work illustrates the potential of LLM-based systems to support evidence-based decision making through more accessible, transparent, and integrated interaction with diverse data sources. With the current software ready for deployment, the next step is therefore not to establish technical feasibility, but to transition the validated prototype into operational use and progressively expand its capabilities. As these technologies mature, they can contribute to more effective use of existing data in conflict early warning, prevention, and crisis management.

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## List of Abbreviations and Definitions

| Abbreviations | Definitions |
|---|---|
| ACLED | Armed Conflict Location and Event Data Project |
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| CCVI | Climate Change Vulnerability Index |
| CoRA | Conflict Risk Analysis |
| EC | European Commission |
| EEAS | European External Action Service |
| ETL | Extract, Transform, and Load |
| FPI | Service for Foreign Policy Instruments |
| FPI.2 | Crisis Response, Conflict Prevention and Peace Building Unit |
| GPT | Generative Pre-trained Transformer |
| INFORM | Inform Risk Index |
| JRC | Joint Research Centre |
| LLM | Large Language Model |
| NUTS | Nomenclature of Territorial Units for Statistics |
| PCM.2 | Peace, Security and Defence Partnerships Division |
| PRIO | Peace Research Institute Oslo |
| RAG | Retrieval-Augmented Generation |
| SQL | Structured Query Language |
| UCDP GED | Uppsala Conflict Data Program Georeferenced Event Dataset |
| UI | User Interface |

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